Category

Curve-Based and Empirical Fixed-Income Risk Measures

BOND RISK AND RETURN USING CURVE-BASED DURATION AND CONVEXITY

Learning Outcome

- calculate the percentage price change of a bond for a specified change in benchmark yield, given the bond's effective duration and convexity

As we have seen, effective duration and effective convexity are curve-based metrics that are useful for gauging the interest rate risk of complex instruments—such as those with embedded contingency provisions—whose future cash flows are uncertain. These metrics are typically determined from bond prices derived using an option valuation model, given specific (parallel) changes in the underlying benchmark government yield curve.

Just as with yield-based interest rate risk measures, effective duration and effective convexity can be used to estimate the percentage change in a bond's full price for a given shift in the benchmark yield curve ($\Delta Curve$), as shown in Equation 3.

$$\%\Delta PV^{Full} \approx (-EffDur \times \Delta Curve) + \left[\frac{1}{2} \times EffCon \times (\Delta Curve)^2\right]. \qquad 3$$

Let's return to BRWA's five-year, 3.2% (semiannual) coupon bond priced at par for settlement on 15 October 2025 and maturity on 15 October 2030. Using Equation 3 and previously derived effective duration and effective convexity,

EffDur = 4.816 and *EffCon* = 26.723,

the percentage changes in the bond's full price for ±100 bp shifts in the benchmark government par curve are estimated as follows:

$\%\Delta PV^{Full} \approx (-4.816 \times 0.01) + [½ \times 26.723 \times (0.01)^2] = -4.68\%$.

$\%\Delta PV^{Full} \approx (-4.816 \times -0.01) + [½ \times 26.723 \times (-0.01)^2] = 4.95\%$.

These bond price changes compare with the previously derived estimates of –4.47% and 4.71% for this BRWA bond when using *ApproxModDur* alone and *ApproxModDur* and *ApproxCon*, respectively, with yield-to-maturity changes of ±100 bps.

As noted earlier, the differences in results using the curve-based measures versus the yield-based interest rate risk measures are due to slightly different bond prices (PV_+ and PV_-) and, therefore, *EffDur* and *EffCon* metrics (versus *ApproxModDur* and *ApproxCon* metrics) since a parallel shift in the government par curve produces a non-parallel shift in spot (zero) rates and one-year forward rates.

Curve-based interest rate risk measures are important for both issuers and investors. For example, as noted previously, for a callable bond, when the benchmark curve shifts downward and interest rates are low relative to the bond's coupon rate, effective duration declines and effective convexity may even turn from positive to negative. From an investor's perspective, these results severely limit the bond's price appreciation in a declining interest rate environment. From the issuer's perspective, the bonds can be bought back at the call price and refinanced at lower market rates.

As for a putable bond, when the benchmark curve shifts upward and interest rates are high relative to the bond's coupon rate, both effective duration and effective convexity decline (but effective convexity always remains positive), which, from an investor's perspective, limits the bond's price depreciation in a rising interest rate environment. This is because the investor can sell the bonds back to issuer at the put price (typically par), rather than at a discounted price, and reinvest the proceeds at the higher prevailing market rates.

more complex securities, such as callable bonds issued by corporations and asset-backed securities (ABSs), include assumptions about the behavior of corporate issuers or the borrowers whose loans make up the underlying ABS assets under different interest rate scenarios. For example, a corporate issuer's decision to call a bond and refinance depends in part on issuer-specific credit spreads. For mortgage borrowers, changes in the value of the underlying asset financed will affect the decision to refinance a mortgage and therefore call an MBS bond. Therefore, estimates of interest rate risk using effective duration and effective convexity are not necessarily improved by choosing a smaller change in benchmark rates. Curve-based interest rate risk measures have become an important tool in the financial analysis of not only traditional bonds but also financial liabilities.

EXAMPLE 2

Using Effective Duration and Convexity to Estimate Bonds' Price Changes Given Benchmark Rate Changes

Continuing from Example 1, to bolster your recommendation that the portfolio manager (PM) not buy the callable bond trading at 101.060, you want to demonstrate the price impact of the bond's negative effective convexity. You assume the benchmark government par curve increases and decreases by 100 bps and use Equation 3, with *EffDur* and *EffCon* estimates based on bond prices from your option valuation model, to derive the following results:

- $\%\Delta PV^{Full} \approx (-7.601 \times 0.01) + [\frac{1}{2} \times -285.168 \times (0.0100)^2] = -9.03\%$.
- $\%\Delta PV^{Full} \approx (-7.601 \times -0.01) + [\frac{1}{2} \times -285.168 \times (-0.0100)^2] = 6.17\%$.

In discussing these results with the PM, you conclude that the negative effective convexity adds to the bond's price decline when benchmark rates increase and subtracts from the bond's price rise when benchmark rates decrease. If the PM's forecast is an equal probability that the benchmark par curve will increase or decrease, the PM would not invest. However, if the PM's forecast is that the probability for the benchmark decreasing is greater, such that expected gain is greater than zero, she may decide to invest in the bond.

Question Set

Questions 1–5 relate to the following information:

As a fixed-income analyst, you are asked to consider the impact of the benchmark government par curve shifting upward by 200 bps on two bonds, as part of an internal stress scenario. From a bond pricing model, the following information on effective duration and effective convexity were obtained.

Bond A: Effective duration = 9.369; Effective convexity = –353.752.

Bond B: Effective duration = 8.517; Effective convexity = –321.756.

1. Calculate the estimated percentage change in the full price of bond A for the 200 bps curve shift.

   Hide Solution

   Solution:

$$\%\Delta PV^{Full} \approx (-EffDur \times \Delta Curve) + \left[\frac{1}{2} \times EffCon \times (\Delta Curve)^2\right].$$

2. Calculate the estimated percentage change in the full price of Bond B for the 200 bps curve shift.

Hide Solution

Solution:

$$\%\Delta PV^{Full} \approx (-EffDur \times \Delta Curve) + \left[\frac{1}{2} \times EffCon \times (\Delta Curve)^2\right].$$

$$\%\Delta PV^{Full} \approx (-8.517 \times 0.02) + \left[\frac{1}{2} \times -321.756 \times (0.02)^2\right].$$

$\%\Delta PV^{Full} \approx -23.47\%$.

3. Based on Questions 1 and 2, if both bonds were in a portfolio and a decision to sell one of them had to be made in light of a forecast of sharply rising interest rates, recommend Bond A or Bond B to sell. Justify your answer.

Hide Solution

Solution:

Sell Bond A since it has the more negative percentage change in the full price of the bond. For every 200 bps increase in the benchmark curve, the full price of the bond drops 25.81%.

4. If you were forecasting the benchmark yield curve to shift downward by 200 basis points, recommend Bond A or Bond B to add to the portfolio in anticipation of the curve shift. Justify your answer.

Hide Solution

Solution:

Add Bond A, since when you calculate the percentage change in the full price for Bonds A and B, Bond A has the higher positive percentage change.

$\%\Delta PV^{Full}$ Bond A $\approx (-9.369 \times -0.02) + [\frac{1}{2} \times -353.752 \times (-0.0200)^2] = 11.66\%$.

$\%\Delta PV^{Full}$ Bond B $\approx (-8.517 \times -0.02) + [\frac{1}{2} \times -321.756 \times (-0.0200)^2] = 10.60\%$.

5. Explain why the estimated percentage change in the price of the bond from a 200 bps upward shift in the benchmark yield curve is more than twice the estimated percentage change in the price of the bond from a 200 bps downward yield curve shift.

Hide Solution

Solution:

Because effective convexity is negative, an upward shift in the yield curve results in a higher absolute price change than a downward shift in the yield curve. For the upward shift, the duration contribution is positive and convexity contribution is negative, offsetting each other.

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